Q2. Domain registration and website technologies

Original question: Perform reconnaissance on an authorized domain to obtain publicly available domain registration details and identify the technologies used by its website. Record the observations and explain the significance of the information collected.

Aim

Collect public registration information and identify technologies used by an authorized website.

Requirements

An authorized domain and URL; whois, WhatWeb, internet access and a writable results directory. Web fingerprinting makes requests to the site. References: [R1], [R2].

Procedure

Step 1 - Set scope and record time

```
mkdir -p "$HOME/EHF_Exam/Q2"
cd "$HOME/EHF_Exam/Q2"
DOMAIN='<AUTHORIZED_DOMAIN>'
URL='https://<AUTHORIZED_DOMAIN>/'
date -Is | tee collected_at.txt
whois --version
whatweb --version
```

Replace the two placeholders with examiner-approved values. `tee` displays output and saves it. Domain and URL are separate inputs because WHOIS does not accept a scheme or path.

Step 2 - Collect registration data

```
whois "$DOMAIN" | tee whois.txt
```

Read the complete response for registrar, registration/expiry dates, domain status and name servers. Registrant fields may be redacted. Some registries now provide registration data principally through RDAP; if WHOIS is unavailable, use the registry's official RDAP/ICANN lookup and save its response, URL and collection time. Absence of public contact details is not suspicious by itself.

Step 3 - Fingerprint the approved website

```
whatweb -a 1 --log-verbose=whatweb.txt "$URL"
sed -n '1,100p' whatweb.txt
ls -l whois.txt whatweb.txt collected_at.txt
```

`-a 1` selects low-aggression fingerprinting. The verbose log retains plugin observations. Inspect redirects: if a destination lies outside approved scope, obtain the correct in-scope URL and do not continue fingerprinting that destination. TLS, HTTP status, title, cookie names, server headers and framework clues may be visible. Missing banners do not prove software is absent.

Expected output - illustrative summary

```
Registrar: Example Registrar
Name Server: NS1.EXAMPLE.NET
HTTP status: 200
Detected clue: HTML5
Server header: nginx (version not disclosed)
```

These are invented teaching values, not registration records or a WhatWeb transcript. Record the literal tool output from your domain.

Interpretation, observation and result

Use a record with columns: source/tool, exact field, actual value, significance, limitation. A registrar identifies the registration provider, not necessarily the host. Name servers identify DNS delegation. An expiry date helps review renewal risk; verify it with the authoritative registration source. Framework and server clues help inventory the website, but banners can be suppressed or altered. Public registration and website information have been collected when the saved records exist and their meaning has been explained; do not claim a vulnerability based on a fingerprint alone.

Common errors

- WHOIS timeout/redaction: record the limitation and use an official registration-data lookup if available.
- No WhatWeb detections: confirm URL, DNS, certificate and connectivity; a generic page may reveal little.
- 403/429: respect the restriction; do not bypass or increase request intensity.

Viva questions

- Is WhatWeb passive? No; it sends HTTP requests to the site.
- What does WHOIS provide? Registry/registrar registration data, with availability and privacy limits.
- Does a server header prove the exact version? No; it is a clue requiring verification.
- Is the registrar necessarily the web host? No; registration, DNS and hosting can use different providers.